

CONTACT

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Room DS19, IIT Roorkee
Uttarakhand, India

EDUCATION

IIT Roorkee
B.Tech in Metallurgy & Materials Science
Expected Grad: 2030

JEE Advanced Rank: 7556

JEE Mains %ile: ~99.5

Grade 12: 94%

Grade 10: 92%

TECHNICAL SKILLS

C++ | Python | Arduino | GPS/GSM Modules | Sensor Integration | Data Analysis

INTERESTS

Speedcubing: Advanced PLL algorithms

Debate: Exploring controversial topics

Soumay Sachdeva

Engineering Undergraduate | IIT Roorkee

Highly motivated engineering student at IIT Roorkee with a strong foundation in analytical problem-solving, embedded hardware, and software development. A dedicated learner who thrives outside their comfort zone, combining technical aptitude with a proven track record of leadership in student administration and event coordination.

PROJECTS

Arduino-Based Smart Backpack — **Hardware/Embedded**

- Designed and prototyped an embedded hardware setup for a smart backpack aimed at student safety and daily utility.
- Integrated a GY-GPS6MV2 GPS module and GSM connectivity to enable real-time location tracking and cellular communication.
- Engineered a centralized breadboard schematic incorporating an LCD display, temperature sensors, and an SOS emergency alert system.

Advanced Physics Analysis — **Core Science**

- Conducted high-precision laboratory experiments, including measuring Planck's constant using color filters via the Photoelectric Effect.
- Successfully executed the Franck-Hertz experiment utilizing argon gas, calculating exact error percentages and analyzing the quantization of atomic energy levels.

LEADERSHIP & EXPERIENCE

Multi-Year School Prefect & Event Coordinator
High School

- Served as a multi-year school prefect, maintaining discipline and actively representing the student body.
- Organized and managed major school events, demonstrating consistent reliability, public speaking, and team management skills.

Academic Presenter
HSI-101 Soft Skills

- Structured and delivered a highly acclaimed presentation analyzing the ethical implications of Valery Legasov's monologue from the miniseries *Chernobyl*.
- Developed persuasive scripts addressing the historical and scientific "cost of lies."